

Rahnuma

रहनुमा رہنما

A working guide to the ILM, PANINI and Zistgah stack, from bytes and scripts to compilers, Pāṇini and quantum substrates

प्रकृति

	written	Devanagari hub	Romenagri	and back
Devanagari	प्रकृति	→ प्रकृति	→ prak_ri_ti	→ प्रकृति
Telugu	ప్రకృతి	→ प्रकृति	→ prak_ri_ti	→ ప్రకృతి
Bengali	প্রকৃতি	→ प्रकृति	→ prak_ri_ti	→ the same identity
Urdu, as usually written	ہندوی	→ हनदवी	→ _hana_dawee	→ हनदवी
Urdu, with zer and sukun	ہندوی	→ हिन्दवी	→ _hin_dawee	→ हिन्दवी

One identity, many scripts. Telugu and Bengali meet Devanagari in one hub, become the same Romenagri word, and come back exactly. Urdu meets the hub at what is written: without its short-vowel marks the word reads hanadawee; with zer and sukun it is hindawi. Every cell here was produced by the Hindawi tools while this guide was built.

Abhishek Choudhary, AyeAI. Version 2.0.0, 3 October 2026.

Rahnuma, the edition for the field: Health and welfare

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Part I: Orientation

1. How to use this guide

In plain words. This guide explains a large body of work on computers, languages and intelligence. You can read it, run the programs on your own computer, or ask an AI to help you, and every example in it was really run.

This guide is for anyone who wants to use, check or extend the work gathered in the *zistgah*, *project-ilm* and *hindawiai* organisations on GitHub: students preparing for GATE, engineers who write compilers, firmware or chip layouts, linguists who work with scripts and sound, and researchers in quantum computing and AI. It asks for curiosity and a laptop, and nothing else.

There are four ways in.

- **Understand.** Part II gives the whole architecture first: where the work comes from, how the parts compose, each term with its nearest analogue and the difference, and the three Acts.
- **Read.** Parts IV and V explain the ideas the work rests on, with the mathematics written out: how numbers and letters sit in memory, what makes a transliteration reversible, how Pāṇini's grammar computes, how a compiler turns text into machine instructions, and how an oscillator network and a quantum circuit attack the same problem.
- **Run.** Part III gives the exact commands to set up Linux, Windows with WSL, or macOS, and to run every component of the estate on your own machine.
- **Ask.** Chapter 24 shows how to put any AI assistant to work on these repositories, and how to check what it tells you.

Conventions used throughout:

- Every example that could run while this guide was being built did run, and the text shows its real output, marked *Executed while this guide was built*. The package's own `gate`, `bash ops/verify.sh`, runs those examples again and fails if any output differs from what is printed here.
- Status words are used exactly. **Tested:** a test in the repository asserts it. **Executed:** it ran and its output is shown. **Written:** the code exists but has not been run here. **Mocked:** a stand-in replaces hardware that is not attached. **Planned:** described, not built.
- Everything said about the repositories was retrieved from the repositories themselves on 29 September 2026: their listings on GitHub, their `CITATION.cff` files and their READMEs. Where a repository does not record something, this guide says so instead of filling the gap.
- A dagger, †, marks a statement that follows the author's working notes and write-ups as supplied on 29 September and 3 October 2026, rather than the repositories.
- Sanskrit is transliterated in IAST. The spelling is British.

The work lives at github.com/zistgah, github.com/project-ilm and github.com/hindawiai, on the site zistgah.org, and in the Zenodo records listed in Chapter 12.

3. Start here: find your way in

In plain words. Four steps to find your part of the guide: who you are, your board or field, what you want to do, and your path with a PDF made for you.

This guide serves a school student and a senior researcher; a village health worker and an investor. Nobody needs all of it. Four steps find your part, on the site or with this page:

1. **Who you are.** A school student, a college student, a postgraduate or doctoral researcher; a teacher; an academic researcher; a working professional, someone in public service, policy or law, someone in community, NGO or village work, an artist, writer or maker; an industry partner or investor; or an AI agent working for a person.
2. **Your board or your field.** A school student chooses a board, ICSE, ISC, CBSE, IGCSE, IB, a state board or another; and a class. Everyone else chooses a field from the international standard classification of fields of study, ISCED-F 2013, the same one FAKIR's lattice uses. It places each subject where its own scholars would: political science is under social and behavioural sciences, not the humanities; law is under business, administration and law; linguistics is under languages.
3. **What you want to do.** Learn the basics, run the software, study or research, teach, contribute, or adopt, fund or invest.
4. **Your path and your edition.** The site lists your chapters in order, with reading times and the components to try first, and offers a PDF edition made for you. Every edition below is a printed selection from this one guide, so all of them stay in step.

Edition	PDF
School student	role-school.pdf
College student	role-college.pdf
Postgraduate or doctoral researcher	role-pg.pdf
Teacher or lecturer	role-teacher.pdf
Academic researcher	role-researcher.pdf
Working professional or engineer	role-professional.pdf
Public service, policy or law	role-public.pdf
Community, NGO or village work	role-community.pdf
Artist, writer or maker	role-creator.pdf
Industry partner, investor or funder	role-industry.pdf
Generic programmes and qualifications	field-00.pdf
Education	field-01.pdf
Arts and humanities	field-02.pdf
Social sciences, journalism and information	field-03.pdf

Edition	PDF
Business, administration and law	field-04.pdf
Natural sciences, mathematics and statistics	field-05.pdf
Information and Communication Technologies (ICTs)	field-06.pdf
Engineering, manufacturing and construction	field-07.pdf
Agriculture, forestry, fisheries and veterinary	field-08.pdf
Health and welfare	field-09.pdf
Services	field-10.pdf

The first public version of this guide, written for advanced researchers and senior professionals, stays as it was at [version 1](#).

Part II: One architecture

8. The ecosystem, term by term

In plain words. A dictionary of the special words used here, each with the nearest everyday idea and what makes it different.

Each entry gives what the thing is in this architecture, the nearest analogue a newcomer is likely to reach for, and the exact difference. The analogue is a door into the idea, never a definition of it. The state in brackets says what exists: established (historical record), released (on the forge, with its DOI where minted), executed (run for this guide), under construction, specification, or planned. Entries marked † follow the author's working notes and write-ups as supplied on 29 September and 3 October 2026, rather than the repositories.

Language and identity

ILM, Integrative Linguistic Multiscript (released; ilm.codes live). *In the architecture:* the linguistic pillar and the identity layer for language, with script, language and standard kept as separate axes, and layers for phonology, transliteration, orthography, lexicon, syntax and semantics, and interfaces, supported by data, validation, development tools, language specifications and a language server; its registry lists 7,867 languages and 226 scripts. *Nearest analogue:* localisation, an NLP toolkit, a transliteration library. *Difference:* it keeps the three axes apart, so that any language can travel in any script, and it gives every script a reversible path through the whole computing stack, down to the symbol table.

Romenagri (established 2003; released; executed). *In the architecture:* the reversible kernel between scripts and ASCII-7. Its paper states six constraints met together: ASCII-7 closure, case independence, diacritic independence, legality as a C identifier, reversibility and linear time, over the Brahmi, Perso-Arabic and Northwest Semitic families, with a canonical-form layer and an ASCII-reduction layer. *Nearest analogue:* romanisation, such as IAST, ISO 15919 or ITRANS. *Difference:* its output is a valid identifier for every tool from lexer to debugger, and it inverts, so a name survives compilation and comes back.

HPS, the Hindawi Programming System (established 15 August 2004; released; executed). *In the architecture:* nine shailis over nine host languages, from assembly, lex and yacc to C++, Java, Python, BASIC and LOGO; a script layer, a language layer and the host standard, composed as transducers over unmodified host toolchains; now HPS's Indic projection under ILM, with an Urdu edition beside it. *Nearest analogue:* a localised programming language. *Difference:* a complete systems stack, not a keyword translation, with names carried through Romenagri.

tajziya (released, doi:10.5281/zenodo.22982238). *In the architecture:* the parser federation for the classical languages, by family, with the author's Sanskrit parser; the front end of the PANINI stack beside ILM. *Nearest analogue:* NLP parsers. *Difference:* organised to feed PANINI's construct model and AAB's paintings.

Computation

PANINI (several members released; some under construction). *In the architecture*: the construct model at the core, with ILM and tajziya in front, PANINI's own front ends and language as middleware, and realization backends behind; it runs through all three Acts. Its family includes the prompt-cycle language in which the cyclers are written (zistgah/panini), the general-purpose PANINI of the merged release (zistgah/humanesque), and the realization arms. *Nearest analogue*: a compiler, a programming language, an intermediate representation. *Difference*: programming languages, prompt cycles and physical, biological and quantum realizations are all projections of one construct model.

PANINIQ (released, doi:10.5281/zenodo.23020648; executed). *In the architecture*: a realization backend: an oscillator substrate, the PEDLER engine and quantum validation. *Nearest analogue*: a quantum simulator. *Difference*: the substrate is an oscillator Ising machine with a learning automaton on it; the quantum circuit validates it rather than replacing it.

PANINIB and **PANINIPHY** (released research platforms, version 1.0). *In the architecture*: PANINIB, the biological realization arm, lowers biological intent through semantic, resolution, CISC, RISC and sequence IRs to a realization plan and an execution run; PANINIPHY, the physical realization arm, lowers physical intent through a physical IR and physical-domain engines, down to modular blocks and the selection of parts by force, stroke and voltage. *Nearest analogue*: computer-aided design, genome design tools. *Difference*: both share PANINI's common spine, language, verification, provenance and ArtifactGraph, adding only their own domain IR and engines; laboratories and machines are external adapters, and neither claims a result its evidence does not show.

Cognition

PEDLER and **QEDLER** (established November 2001; record doi:10.5281/zenodo.17497559; QEDLER a research framework). *In the architecture*: the point-event-driven learner; a six-tuple extension of the Turing machine over balanced ternary; the source of the primitives Act (executable intent), Inclination (a signed directional gradient on transition edges) and Natural Justice (a gate on the passage from intent to Act); QEDLER extends it to physics. *Nearest analogue*: an event-driven or neuromorphic learning model. *Difference*: it predates the present generation of models and carries the executable-intent semantics the constitutional corpus is built from.

PRATIK and **Zamin** (released: pratik_core_mvp doi:10.5281/zenodo.21288232, zamin doi:10.5281/zenodo.21297556). *In the architecture*: PRATIK, Participatory Recursive Adaptive Trans-Intelligence Kernels, on the divider between Acts I and II, with a kernel in C++ and CUDA; Zamin, its physical balanced-ternary substrate with a poised zero. *Nearest analogue*: a neuromorphic kernel on novel hardware. *Difference*: balanced ternary and event-driven from the logic up.

CEM, with CEMb, CEMs and Eco-CEM (specification; Act II). *In the architecture*: cognitive enablement modules over one kernel, $\mathcal{E} = (\mathbf{E}, \mathbf{C}, \mathbf{\Pi}, \mathbf{W})$, realised on biological substrates (BCI, prosthetics, closed-loop cognition, operated through Interglial Healthcare), on synthetic ones (processors, distributed systems, robotics) and on ecological ones. *Nearest analogue*: robotics, BCI, prosthetics. *Difference*: one invariant kernel across substrates; the superscript is part of the name.

The AyeAI Triad (specification with formal closure). *In the architecture*: AyeAM = $\langle \mathbf{S}, \mathbf{R}, \mathbf{C} \rangle$ for embodiment, AyeAI = $\langle \mathbf{M}, \mathbf{I}, \mathbf{G} \rangle$ for cognition, AyeCNSe = $\langle \mathbf{T}, \mathbf{Ch}, \mathbf{\Sigma} \rangle$ for coordination and communication across media, with AyeAI at the apex. *Nearest analogue*: a perception, cognition and

action loop. *Difference*: a triad, not a loop and not a pipeline.

Knowledge and agency

UKOP, FAKIR and Dhancha (FAKIR released, doi:10.5281/zenodo.21436550; Dhancha released, doi:10.5281/zenodo.22821645; UKOP a specification). *In the architecture*: UKOP, the reference architecture for human knowledge; FAKIR, its kernel over ISIC, ISCO and ISCED crossed with AGI layers and language; Dhancha, the spine every domain is built on, whose rule is that the engine is common and the workflow is not. *Nearest analogue*: a knowledge graph or reasoning engine. *Difference*: domain invariants are enforced by tests, and resolution and verification are primitives.

Cyclers (six released and minted; Act I). *In the architecture*: matba, khwab, awaz, tilasm, pench and yadein, classified by what they produce, plus genie, each with its own contract, state model and evidence rules, written in PANINI and run by one engine. *Nearest analogue*: an agentic harness or orchestration loop. *Difference*: AI-agnostic by construction, inspected by a person at every step, and the recorded method, never the content, is itself the reproducible work. † An older form of the corpus names SAFAR in the sixth position, where the current corpus has yadein.

Kitab (released). *In the architecture*: the config-driven book template that matba publishes through, and † in the author's notes the persistent artifact and publication layer that keeps an artifact's state, its creative lineage, its execution trace and evaluations, its forks and descendants. *Nearest analogue*: a publishing template. *Difference*: a cycler publishes through Kitab; Kitab is not a seventh cycler.

Research Kundali (released). *In the architecture*: a research map of a person, a lab or a paper, built from an ORCID, a lab name or a DOI ([project-ilm/research-kundali](https://project-ilm.com/research-kundali)); † in the author's notes, the systematic record of claims, artifacts, chronology, provenance, objections, evidence and responses, so that every claim can be checked against its evidence. *Nearest analogue*: a researcher profile. *Difference*: it records evidence and discrepancies, not reputation.

GENIE (released). *In the architecture*: the Generalized Emotive-Narrative Interaction Engine, storyteller, poet and painter, and the research cycle through its Prompt Operating System, with six verbs (create, verify, execute, measure, falsify, integrate), nine epistemic tags and a gate that a simulation cannot pass in place of an experiment. † In the author's own formulation GENIE is itself a cycler, the creative cycler, which can decorate the modality cyclers without erasing their identities. *Nearest analogue*: a creative or research AI agent. *Difference*: governed by a constitution of primitives; orchestration and composition, not an implementation.

AAB (released; Act I). *In the architecture*: the paint program for systems, from painted intent to verified, sealed artifact; the gamified studio of verification-gated quests; home of the process cyclers; the process half of the estate's software factory, with FAKIR as its component registry. In the author's words, AAB and FAKIR together constitute the working definition of AGI used here. *Nearest analogue*: a low-code studio or a software factory. *Difference*: every task is a painting with a manifest, quest stages and oracle-gated verification, and every component keeps its provenance.

Mez, the Cognitive Workbench (released). *In the architecture*: the local-first desk that brings independently existing systems together for an exercise: composition, exercise, observation, synthesis, a new artifact. *Nearest analogue*: an IDE or a notebook. *Difference*: composition without absorption; every composed system keeps its own identity and life.

Method and evidence

VGC, Verification-Gated Human-AI Co-Development (released, doi:10.5281/zenodo.21264248). *In the architecture*: the method by which people and AI build together, with every step gated by verification. *Nearest analogue*: test-driven development or human review of AI output. *Difference*: the gate is a verified artifact, not a reviewer's impression, and the AI's report that something passed is never the evidence that it passed.

COPA, the Cost of Perceived Authority (released, doi:10.5281/zenodo.21782217). *In the architecture*: a protocol for measuring how much an AI system asserts before it verifies, through the Authority Projection Index, with its hypothesis stated beside a fair null. *Nearest analogue*: an AI evaluation benchmark. *Difference*: it measures projected authority rather than accuracy, and its target is calibrated trust.

† One rule binds all of these: a record that has not been retrieved is not thereby false. Unretrieved is a state of the reader's context, not of the world, and a claim moves from unknown to false only on evidence.

Continuity and civilisation

Humanesque (specification; the merged release on the forge). *In the architecture*: the core over which everything above is a typed hypergraph, and the constitutional realm derived from PoIE through recursive sovereignty. *Nearest analogue*: an AGI architecture. *Difference*: a constitutional and epistemic realm as well as a technical substrate, not a product.

Kaivalyik, Zistgah and Cosmopolis (projections). As in Chapter 6: parallel cultural projections of the whole, not stages. Kaivalyik carries Synthematic Pragmatic Realism, SPR, the framework that heads Act III ("Kaivalyik: Towards a Synthematic Pragmatic Realism for the AGI Singularity"); Zistgah carries the habitats and the elements.

TransEg (released, doi:10.5281/zenodo.21321558). *In the architecture*: the transferred alter ego, the mechanism by which a constituted identity continues; a local digital-twin reference implementation; yadein's staggered upload feeds it. *Nearest analogue*: a digital avatar. *Difference*: continuity of an identity under the constitutional corpus, not a persona.

PAT.AL, VIDYA, GramSheel, Project VIKRAM and TWISHA (established). PAT.AL, the Participatory Alliance for Technology, Access and Livelihoods, is participatory infrastructure; VIDYA bridges to AyeAM; Project VIKRAM, Virtualized Infrastructure for Knowledge-driven Rural Ascension Management, runs under the GramSheel Foundation beside TWISHA. They sit on the dividers between the Acts. The question the author puts to them is whether technology can distribute capability without concentrating power.

The reference lab (planned; its launch scripts are the next release). An AGI-capable laboratory for 30 to 40 thousand, able to carry 80 to 90 per cent of the research behind high-end AI, robotics and automation papers, repeated as a pattern for every domain and aligned with GATE-level courses in computer science and in robotics and automation.

Also in the working notes

Modular self-reconfiguring robots †: very simple modular blocks with face-to-face connections and electro-permanent-magnet contacts, assembling into larger structures, ruggedised for decentralised manufacture with low-cost embedded control; PANINIphy's modular-block examples are the compiler side of this. **Further realization families** †: chemical, photonic, and a computational-biological intermediate layer of the AlphaFold class, built as an independent spine rather than a dependency. **The build engine** †: the direction of the whole is a self-reflecting, self-evolving and self-sustaining build engine in which PANINI captures and expresses the construct and the intent, and agency, human or machine, biological or physical, does the building. **INDUS and KHĀK** † are named in the notes among the lineages and the robots, alongside ANGEL, TARA and NI2A2.

10. Three families, set apart: the AyeAI Triad, GramSheel and Project VIKRAM, Humanesque and its projections

In plain words. Three groups explained one at a time: the AyeAI Triad of thinking, body and communication; GramSheel and Project VIKRAM for villages; and Humanesque with the names it takes in each culture.

Three families of work are easy to blur into each other or into the rest of the estate. This chapter sets each apart, with what its own record says.

The AyeAI Triad

The Triad has three members, AyeAI®, AyeAM® and AyeCNSe®, with AyeAI at the apex and AyeCNSe and AyeAM at the base. It is a triad, not a pipeline and not a loop: a loop running from the environment through AyeCNSe, AyeAI and AyeAM back to the environment was proposed elsewhere and rejected. Its formal closure gives each member a tuple:

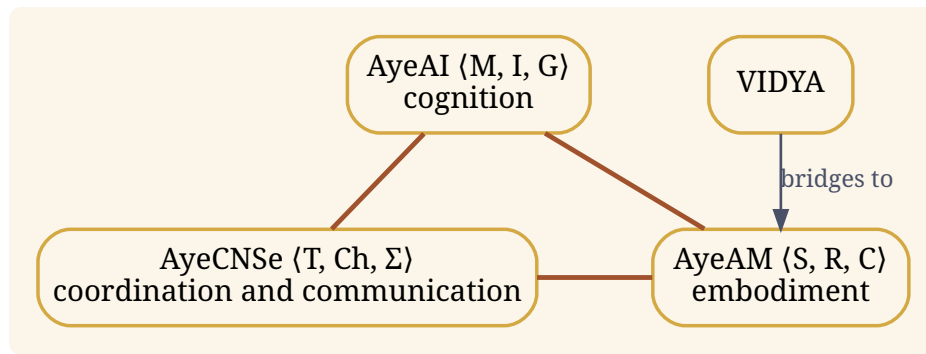


Figure 8. The AyeAI Triad: a triad with AyeAI at the apex, not a loop and not a pipeline.

$$\mathbf{AyeAM} = \langle S, R, C \rangle, \quad \mathbf{AyeAI} = \langle M, I, G \rangle, \quad \mathbf{AyeCNSe} = \langle T, Ch, \Sigma \rangle.$$

AyeAI is cognition, the apex: “the inclusive AI”, in the words of its site, and the organisation [ayei](#) holds its platforms, among them an open public stack for synthetic intelligence (opssi), virtual deployment infrastructure for scientific and cognitive computing (ayevdi), Staggered Upload™ (upload), the AyeAI Singularity Public License (spl) and the NI2A2 work.

AyeAM is embodiment: the body through which intelligence acts, across ground, air, water, space and habitat. VIDYA bridges to AyeAM without being a member of the Triad.

AyeCNSe is coordination and communication across media: the form studio at [forms.ayecnse.site](#) and the Human Context Model paper at [hcm.ayecnse.site](#) are its published faces.

Two repositories mark themselves as PEDLER and AyeAI Triad work: PRATIK’s kernel and Zamin’s ternary silicon.

GramSheel and Project VIKRAM

The GramSheel Foundation is the social architecture of the estate and the older of its lines: † it began as the Nature Welfare Society in 1993. Its principles, HEJLP, are health, education, justice, livelihood and peace. Its Village Knowledge Center (Universal Device) was a Stockholm Challenge finalist in 2010, and TWISHA runs under the same foundation.

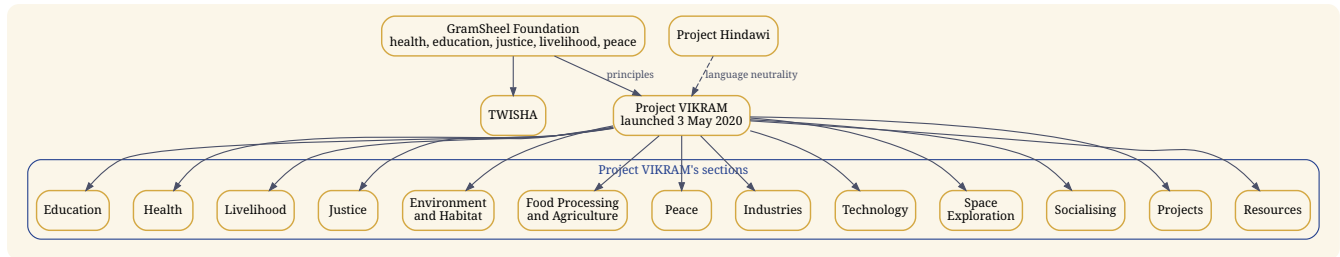


Figure 9. GramSheel and Project VIKRAM, with VIKRAM's sections as its own pages name them.

Project VIKRAM, Virtualized Infrastructure for Knowledge-driven Rural Ascension Management, is † the technical implementation of GramSheel's principles. It was launched on 3 May 2020 as the Project Vikram Journal, “an inclusive rural digitalization platform inspired by the principles of GramSheel”, under the motto गाँव बढ़ेंगे तो सब बढ़ेंगे, when the villages grow, everyone grows. Its own pages (pvjournal.github.io) name its components:

Component	What its page covers
Education	learning and schooling
Health	healthcare, open to submissions as issues, with tools for an accessible virtual hospital
Livelihood	the ISIC domains of economic activity
Justice	social justice
Environment and Habitat	the living environment
Food Processing and Agriculture	agriculture and food
Peace	harmony through social dialogue
Industries, Technology and Space Exploration	the technical frontiers it reaches towards
Socialising, Projects and Resources	the community and its shared material

With Project Hindawi, its page says, VIKRAM works towards language neutrality across technical fields. In the three Acts, GramSheel and Project VIKRAM stand on the divider between Acts II and III.

Gram Swaraj and the questions behind GramSheel and VIKRAM

Gandhi's *Hind Swaraj* (1909) argued that self-rule is more than a transfer of authority, and his *Constructive Programme: Its Meaning and Place* (1941) placed the rebuilding of daily life inside the work of independence: its eighteen items include village industries, village sanitation, basic and adult education, the provincial languages and a national language, economic equality, the kisans, labour, the adivasis and women. The estate does not present itself as a restatement of these ideas. The point of contact is a question that becomes unavoidable once computation is infrastructure: can a community govern itself if the knowledge and the machines its decisions depend on are controlled elsewhere?

On Gandhi Jayanti, 2 October 2026, the author put the questions behind GramSheel, Project VIKRAM and Humanesque in his own words. Among them:

- Can rural communities become producers of knowledge and technology, not merely consumers of systems designed elsewhere?
- Can technology distribute capability without concentrating power?
- Can artificial intelligence strengthen individual and community sovereignty rather than replace it?
- Can knowledge remain plural, multilingual and culturally grounded while becoming computationally executable?
- Can verification, provenance and human agency become constitutional properties of intelligent systems?
- Can GramSheel build the local knowledge infrastructure, can Project VIKRAM turn that infrastructure into practical capability, and can Humanesque provide a larger architecture in which intelligence, identity, agency, provenance and equity remain connected?

His answer to the old division of the world is a direction rather than a side: to march together, not as a Global South or a Global North, but towards a Global Centre, where capability is distributed and humanity remains sovereign.

Humanesque and its projections, with their names

Humanesque is the core, with its site at humanesque.site and its organisation at hmnsq: the shared executable typed hypergraph, the ontology federation, the common execution and evidence spine, and the components every projection reuses. Its constitutional ground is PoIE, from which sovereignty recurses to every level. Its core stem repositories all belong to Humanesque; each projection gives them its own culturally sensitive names and sites.

Kaivalyik, the Indic projection, including the Advaita framing (kaivalyik.org, organisation [kaivalyik](https://kaivalyik.org)): consciousness, intelligence and transcendence; it carries Synthemetic Pragmatic Realism, Kaivalyik AGI and Jyotish. **Zistgah**, زیستگاه, the Persian and Islamicate projection (zistgah.org, organisation [zistgah](https://zistgah.org)): life, planet and ecology, a habitat to live and work in; it carries the elements, the dome and the habitats on Earth (Asli Zistgah), the Moon (Zistgah-e-Mahtab) and Mars (Zistgah-e-Bahram, † conceived earlier as Mangal Kamna). **Cosmopolis**, the Greek, Western and classical projection (cosmopolis.org, organisation c-polis): societies, civilisations, space and deep time.

The stem names below are the projection-neutral names of the components. Today the working repositories live in the zistgah organisation, so most of them carry their Zistgah names; two stems already show a second projection, HPS in Hindawi and CHAKRA in Jyotish.

Stem, in Humanesque	Zistgah	Kaivalyik
The programming system	Urdu edition (urdu-ilm)	Hindawi
The observatory and its calendars	Chakra, the turning sky	Jyotish
The ground: balanced-ternary silicon	Zamin, زمین	
The water: the factory process	AAB, آب	
The air: the environmental replica	Fiza, فضا	
The shared virtual dome	the dome	
The Cognitive Workbench	Mez, میز	
The press	Matba, مطبع	
The cutting room, the visual cyler	Khwab, خواب	
The listening room, the audio cyler	Awaz, آواز	
The immersive cyler	Tilasm, طلسم	
The embodied cyler	Pench	
The record	Yadein, یادیں	
The book template	Kitab, کتاب	
The domain spine	Dhancha, ڈھانچہ	
Communication engineering	Ertabat, ارتباط	
Device control	Zasab, اعصاب	
The door for any AI	Darwaza, دروازہ	
Scan and store	Tasvir, تصویر	
The plates	Lawh, لوح	
The parser federation	Tajziya, تجزیہ	
The guide	Rahnuma, رہنما	
The Geometry of Becoming	Tabdili, تبدیلی	
The Geometry of Living	Zindagi, زندگی	
The Geometry of Persisting	Qaem, قائم	
Topological Bildung	Tarbiyat, تربیت	

Some stems keep one name in every projection: Humanesque, ILM, Romenagri, PANINI with PANINIq, PANINIb and PANINIphy, PEDLER, QEDLER, PRATIK, CEM, UKOP, FAKIR, GENIE, TransEg, PoIE, VGC, COPA, and the provenance systems Tok DOI, spiguard, Candor and Misty DOI. Kaivalyik's and Cosmopolis's own names for the rest are given with their organisations, as each projection takes its repositories.

Appendices

D. For researchers in science, technology, engineering and mathematics

In plain words. A guide for researchers in science, technology, engineering and mathematics: what can be read, run and tested now, and what is still open.

Each field below lists what in the estate can be read, run and tested now, and what is open. Status words keep their meaning from Chapter 1: tested, executed, written, mocked, planned.

Computer science

Programming languages and compilers. Hindawi's composed transducers over unmodified host toolchains; the Romenagri symbol bridge through ELF, DWARF and GDB; the PANINI construct model, with 39 constructs in 27 languages and 216 decorators across 8 host languages; the bootstrap and its self-hosting fixed point (Chapters 21 and 6). Executed here. **Open:** `std2hin` doubles every underscore, so vowel-initial names do not round-trip back from C; the guru shaili emits no `#line` directives; the YACC shaili awaits a fresh conformance run.

Systems. The hinlin kernel tree and the mass conversion of real C sources, including kernel trace code, into Hindi and back (`hindawiai/chintamani`, `Romenagri/mass_hindawi`).

AI and machine learning. The cyclers and GENIE as AI-agnostic, human-inspected method (Chapter 8); Romenagri as a tokenisation substrate, where bytes fall 1.76 times and the alphabet from 51 to 24 symbols without loss, but token counts did not fall at matched merges, so the predicted gain is still to be shown; COPA's Authority Projection Index as a measurable property of a model's behaviour.

Software engineering. Executable contracts and gates in every repository; the provenance chain, Tok DOI, spiguard, Candor and Misty DOI; VGC; the component cards of Chapter 15, each the result of a real run.

Data. ILM's registry of 7,867 languages and 226 scripts; 13,668 keyword standards; the corpora in the Romenagri tree.

Electrical and electronic engineering

The open chip-design flow and the 28 nm tape-out catalogue, jugaad28, with its evidence-gated descriptors (Chapter 22); Zamin's ternary cells in SPICE, whose testbench needs attention on ngspice 42 (Chapter 15); the PRATIK kernel, whose CPU build passes and whose CUDA build needs an NVIDIA toolchain; device control on ESP32, zasab, built but not yet pushed.

Physics

QEDLER, the event-hypergraph research programme for physics, self-declared as a framework with open problems; oscillator Ising machines and Kuramoto dynamics in PANINIQ, with their Lyapunov function (Chapter 23); superconducting qubits and the physics of the cold (Chapter 22).

Mathematics

Coding theory for reversible transliteration: Kraft and McMillan, and Sardinas and Patterson (Chapter 19); formal languages and rewriting in Pāṇini's grammar, with pratyāhāras as interval classes over an ordered list (Chapter 20); graph theory in the typed hypergraph and in Max-Cut; dynamical systems in the order parameter and in the oscillators' energy; the formal tuples of the CEM kernel, $\mathcal{E} = (E, \mathcal{C}, \Pi, W)$, and of the AyeAI Triad.

Biology and medicine

PANINIB, the biological realization arm, which lowers intent to sequence IR and claims no wet-laboratory result; CEMb, operated through Interglial Healthcare; the Quantum Neuromorphic BCI programme, with brain-to-text, cognitive prosthetics and intelligent rehabilitation as directions needing clinical collaboration; VIKRAM's Health section and its tools for an accessible virtual hospital; the earlier HMSEI wearable and the Dr Rho telepresence platform.

Cognitive science and neuroscience

PEDLER's two branches, through LVF and MLCNE to the cognitive enablement modules, and through NI2A2 to ANGEL; its primitives; the brain-computer interface work of Act III.

Astronomy and the earth and space sciences

CHAKRA, the offline observatory and calendars, whose JavaScript and C99 versions produce byte-identical output; Jyotish, derived from it; the habitats, with a lander embodiment on the Moon and a quadroter on Mars inside the shared dome.

Robotics and manufacturing

PANINIphy, the physical realization arm, which compiles physical intent and chooses parts by force, stroke and voltage; pench, the embodied cyler; † the modular self-reconfiguring robots of the notes.

E. For practitioners: teachers, policy makers, communities, makers and engineers

In plain words. A guide for teachers, policy makers, community workers, artists and engineers: where to start and how to take part.

Teachers and lab builders. Start from the reference lab, the GATE chapter and the component cards, which double as laboratory sheets: each says what to install, what to type and what a student should see (Chapters 16, 26 and 15). The reader views let each student follow a path suited to their age and background, and remember what they have read.

Policy makers and administrators. The questions that matter are in Chapter 10 and Appendix C: sovereignty in its testable technical sense, the ability to rebuild, verify, modify and run (Chapter 21); provenance that makes claims checkable; and infrastructure that leaves capability with communities. The executable contracts of the repositories show what verifiable governance of software looks like in practice.

Community workers and health practitioners. Project VIKRAM's sections cover education, health, livelihood, justice, environment and habitat, food and agriculture, and peace; its health section accepts submissions as issues. The Urdu edition and the Brahmi hub show work in people's own scripts; CHAKRA and Jyotish run offline.

Artists, writers and makers. The cyclers produce print, film, sound, immersive work, embodied work and the record; matba turns posters into a sealed book; GENIE writes and paints; PANINIphy turns a physical idea into parts. Every cycle is recorded as method, so a piece of work can be repeated, shared and credited.

Engineers in industry. Adopt one component at a time: read its card, clone it, run it, read its `CONTRACT.md` and its gate, check its DOI and provenance, then integrate it through its interface. † The notes give the full adoption ladder: the canonical artifact, a one-page explanation, the nearest industry analogue and the exact difference, a working demonstration, the repository, the DOI and timestamp, the interface, a case study, and the commercial route.

Contributors. Fork, open an issue, send a pull request. Every repository's gate runs on your change; every claim must point to its evidence; the AI you use is welcome, and its report that something passed is not the evidence that it did.

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